
Event-Driven L2 Replay

An artifact-verifiable BTCUSDT market-making case study

Pranav Pillai

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Frozen research snapshot: 1066950

Repository: github.com/pranavpillaiNUS/l2-mm-system

Scope in one sentence. This repository is a tested Python research system for deterministic limit-order-book reconstruction, event-driven execution simulation, and hypothesis testing. It makes no claim of a profitable live strategy or exchange-validated production execution.

Central finding. In the frozen 24-window development study, normalized order-flow imbalance was strongly associated with one-second forward mid-price movement at the population level (HAC $t = 64.63$, effect $+0.123$ bps per signal standard deviation), but the pre-specified fill-conditioned screen failed at both queue-model endpoints. The conditional bucket pattern was non-monotone, so the proposed OFI-gated maker was not advanced and no candidate strategy was evaluated on the holdout.

Model-version boundary. All economic numbers in this report are frozen historical evidence produced by `legacy_book_update_v1`. The current Python behavioral reference, `event_driven_v2`, has passed implementation acceptance but its development-panel economic rerun is pending. The old and current models are deliberately not conflated.

Venue and instrument: Binance spot BTCUSDT. L2 research period: April–May 2026. Software: Python 3.11. The raw capture is not distributed, canonical derived artifacts and integrity identities are committed.

Abstract

This report documents the design, failure analysis, and present state of a market-microstructure research system. The system records Binance spot depth and trades, reconstructs the L2 book with decimal arithmetic, merges events deterministically, simulates post-only passive orders with modeled queue position and latency, and produces auditable P&L, markout, queue, and signal diagnostics. An earlier hourly bar backtester established reusable interfaces but is retained only as an educational precursor because its same-close execution and return annualization are too coarse for rigorous strategy evidence.

The frozen development study used 24 non-overlapping five-hour windows and two endpoints for queue cancellation credit. The passive microprice baseline did not demonstrate stable positive economics. Normalized one-second OFI had a strong, monotone population association with forward mid drift, but its pre-specified conditional comparison was only +0.127 bps under proportional credit and -0.376 bps under no credit, below a 1.0 bps screening threshold. Because the full conditional dose-response was non-monotone, the evidence did not justify advancing the candidate. No candidate strategy was evaluated on the 12-window holdout. Only its pre-strategy regime descriptors were examined.

A subsequent audit found that the historical simulator quantized private order arrival to book updates and applied cancels immediately. Rather than relabel the old results, the project froze them under `legacy_book_update_v1` and built `event_driven_v2`, with exact scheduling on the model's millisecond clock, post-response-gated snapshot recovery, cancel/fill races, own-order FIFO, trade-volume conservation, fail-closed gaps, and a hard working-exposure envelope. At the 4 August 2026 publication checkpoint, a fresh clone reports 376 deterministic tests passed and three optional raw-data smoke checks skipped under Python 3.11. All 379 pass when the selected captures are restored. The live recorder check is a separate manual network operation. Economic results under the current model remain pending. The project contribution is therefore the combination of a transparent negative/conditional development result, explicit model-risk boundaries, and an artifact-verifiable Python behavioral reference for later performance work.

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1 Executive Summary

1.1 Research question

The project asks a narrower question than “can an order-book signal predict the next price move?” It asks whether a signal remains useful after the strategy’s own execution process selects which observations become passive fills. That distinction is central to market making. A resting limit order is most likely to trade when an aggressor reaches it, and that event can be informative about the next move. A population-level forecast can therefore be real while the subset of states in which the maker is filled remains toxic. This is the adverse-selection problem described in classical microstructure models such as Glosten and Milgrom [5], and OFI is motivated by the empirical link between order-book events and short-horizon price impact in Cont et al. [3].

The system was built to measure that gap between *signal association* and *execution-conditioned economics* without hiding assumptions about queue position, event ordering, fees, or missing data.

1.2 What was built

The current repository contains four connected capabilities:

- a bar-level backtesting scaffold retained as project history,
- spot and perpetual-futures data recorders, with spot L2 and trade data forming the completed research dataset,
- a deterministic event replay and execution simulator with post-only behavior, configurable latency, partial fills, L2-derived FIFO queue position, and maker/taker fee assignment, and
- a research pipeline for integrity screening, panel selection, bootstrap intervals, OFI diagnostics, model stress, decision gates, artifact hashing, and a strategy-sealed holdout.

The most important engineering work occurred after the historical study. Model audits showed that the old simulator activated private orders only on the next book update, made cancellations immediate, and applied REST snapshot state at a local tag captured before the request completed. The current `event_driven_v2` implementation instead schedules order and cancel arrivals on the replay clock, gates legacy snapshot recovery on a post-response receipt proxy, models cancel/fill races, conserves recorded trade volume across overlapping own orders, and refuses to write into the frozen result root.

1.3 Evidence and decision

The frozen development experiment covers 24 non-overlapping five-hour windows. The passive microprice baseline was negative in 18 of 24 windows at each tested queue endpoint. The proportional-credit mean was -1.043 USDT per window with a 95% window-bootstrap interval $[-2.332, -0.032]$. The no-credit mean was -0.686 with interval $[-1.883, +0.290]$.

Normalized OFI was much stronger than microprice as a population signal. At one second, the pooled association had HAC $t = 64.63$, $R^2 = 0.0366$, and an effect of +0.123 bps per signal standard deviation. All 24 per-window coefficients were positive, and the six unconditional bucket averages were monotone.

The pre-specified conditional screen nevertheless failed. It compared the most-populated adverse side-aligned bucket with the most-populated nonnegative bucket at a 30-second horizon. The separation was +0.127 bps under proportional queue credit and -0.376 bps with no credit, below the 1.0 bps screen. The full bucket pattern was non-monotone, including a smaller intermediate bucket with a more favorable mean. Consequently, the correct conclusion is not that OFI is universally useless after conditioning. It is that the defined conditional gate did not yield a robust, monotone separation strong enough to justify a candidate run. The `OFIGatedMM` candidate was blocked and no holdout strategy result was generated.

1.4 Why the negative result matters

The project rejected several opportunities to report an attractive but weakly supported result:

- Early apparent profitability contained taker fills created by latency, post-only enforcement removed that interpretation.
- Snapshot synchronization and stale-diff issues were corrected and the affected study was regenerated.
- A strong unconditional OFI statistic was not treated as executable edge after the fill-conditioned gate failed.
- A candidate was not tried on the strategy-sealed holdout after failing its development premise.
- A later execution-timing flaw created a new model and artifact namespace instead of being retroactively attached to old results.

For a research record, this audit trail is more informative than a fragile positive backtest. It demonstrates the ability to define an estimand, identify where simulation assumptions enter it, reject contaminated evidence, and stop an experiment when the protocol says to stop.

1.5 Version map

The project has similarly named study and model versions. Table 1 is the authoritative map.

Table 1: Study and execution-model version map.

Label	Meaning	Status and permitted claim
Study V1	Six-window exploratory study	Frozen and superseded by the larger historical study.
Artifact V2	24-window Phase A/B/C development study	Frozen historical evidence, generated at research commit 1066950.
Legacy model	Historical private-timing and queue semantics	Superseded, the source of V2 execution-derived numbers.
Current model	Current Python behavioral reference	Mechanics accepted locally, no development-panel economics yet.
Artifact V3	Results produced by <code>event_driven_v2</code>	Pending, the development rerun has not been executed.
Holdout	12 selected five-hour windows	Strategy-sealed, no candidate strategy output exists.

Current public interpretation. The repository is ready to demonstrate the engineering reference and the discipline of the frozen case study. It is not yet ready to claim that the historical economic conclusion survives the current execution model. The V3 development rerun is the next research milestone.

2 Project Evolution

2.1 Phase 1: an educational backtesting scaffold

The project began with BTCUSDT hourly OHLCV bars. Four simple strategy families were implemented behind a common interface: momentum, mean reversion, moving average crossover, and Bollinger bands. The surrounding framework separated configuration, data loading, portfolio accounting, transaction costs, metrics, plotting, parameter sweeps, and walk-forward orchestration.

That phase produced valuable engineering lessons: fees could erase a gross edge, stateful strategy instances had to be recreated between runs, expanding window selection changed parameter rankings, and

orchestration code required as much testing as individual components. Those lessons directly informed the L2 design.

The historical Phase 1 performance table is *not* treated as rigorous strategy evidence in this report. A later publication audit identified four limitations:

1. strategies observe the current bar close and execute at that same close,
2. hourly equity observations were annualized with 365 periods per year,
3. the displayed “OOS Sharpe” is a mean of monthly Sharpe estimates rather than the Sharpe of one stitched OOS return series, and
4. the historical optimizer source was overwritten by the plotting pass. A runner has since been reconstructed from the logged grids, but the original performance artifacts are not promoted to current evidence.

The current metrics helper now infers the observation frequency from its timestamp index, and a regression test covers the hourly-versus-daily scaling. The historical Phase 1 CSVs were not rewritten, so their published ratios retain the original limitation.

These issues do not affect the independently implemented L2 replay. They do mean that Phase 1 should be read as an engineering precursor and a record of learning, not as evidence that any bar strategy generalizes. This distinction is now made explicit rather than allowing precise-looking ratios to imply a stronger study.

2.2 Why bars were insufficient

Market making depends on phenomena that bars remove:

- whether a limit order was passive when it reached the exchange,
- volume ahead at a price and how trades advance queue position,
- partial fills and residual inventory,
- entry and cancellation latency,
- event ordering when book changes and trades share a timestamp, and
- whether a fill is followed by a favorable or adverse price move.

The project therefore shifted from optimizing directional bar signals to reconstructing a deterministic L2 event path and treating execution as part of the hypothesis.

2.3 Phase 2: replay, execution, and a frozen study

Phase 2 added data recorders, snapshot-aware parsers, an L2 book, depth/trade merging, execution simulation, passive quoting strategies, and markout/P&L analysis. Exploratory work selected a microprice reference, a 2 USDT half-spread, and a five-second requote interval. Six anchor windows were later retained inside a larger 24-window development panel. Because those anchors were involved in exploration, the final panel is broader and pre-selected but not a wholly untouched confirmation sample. The report preserves that fact.

The expanded study was organized as:

Phase A remeasure the fixed passive baseline at queue-credit endpoints,

Phase B test whether OFI satisfies a population and fill-conditioned premise gate before running a strategy that uses it, and

Phase C stress queue-cancellation credit and entry latency, then close the development arc if no candidate qualifies.

No candidate qualified. No candidate strategy was evaluated on the holdout.

2.4 Phase 2.5: execution-model closure

After the study was frozen, review of private-event timing found that the historical engine used the next depth event as the activation point for an order whose modeled arrival might have occurred earlier. Cancels were immediate, so cancel/fill races were absent. Overlapping own orders also required a stronger FIFO decomposition to guarantee that one recorded trade could not fill more simulated volume than it contained.

The response was a provenance boundary:

- historical artifacts were labeled `legacy_book_update_v1`,
- the frozen result directory became write-protected to current scripts,
- the new implementation was named `event_driven_v2`,
- current results were assigned a distinct V3 namespace, and
- cross-model comparisons were restricted to descriptive sensitivity, not treated as estimates from the same execution process.

This is the current stage. The Python mechanics are tested, V3 development economics remain pending.

2.5 Timeline and decision discipline

Table 2: Condensed project timeline.

Period	Milestone	Research consequence
Jan–Apr 2026	Bar framework and walk-forward exercise	Established reusable interfaces and exposed the importance of costs and state isolation.
Apr–May 2026	L2 recorder, replay, execution, and exploratory baseline	Profit contaminated by accidental taker fills was rejected after post-only enforcement, a later positive one-window passive result failed broader-window evaluation.
May 2026	Reconciliation, markouts, queue and same-ms audits	Shifted emphasis from headline P&L to fill quality and model risk.
Jun 2026	Frozen 24-window Phase A/B/C study	Baseline remained unattractive, OFI premise gate blocked the candidate, the holdout stayed strategy-sealed.
Jul 2026	Private-event timing and own-FIFO correction	Froze legacy evidence and accepted <code>event_driven_v2</code> as the Python reference.
Aug 2026	Public-release audit and technical report	Narrowed overstrong claims, demoted Phase 1 performance, and separated clone-level verification from full-data replay.
Next	V3 development-panel rerun	Determine whether current-model fills and economics preserve or change the historical conclusions.

3 Data and System Architecture

3.1 Recorded streams

The primary research dataset is Binance spot BTCUSDT. Two independent streams are recorded into hourly gzip-compressed JSON Lines files:

- depth diffs plus periodic 1,000-level REST snapshots, and
- aggregate trades, including exchange trade time, aggregate trade ID, price, quantity, and the buyer-maker flag.

Depth quantity zero removes a price level. A nonzero quantity replaces the aggregate displayed quantity at that level. Snapshots establish a new `lastUpdateId`. Diffs at or before that ID are discarded, and replay resumes only when the first retained diff satisfies the exchange bridge condition [1, 2]. Depth diffs expose Binance event time `E`, aggregate trades expose matching time `T`. REST snapshots have no exchange event timestamp.

The legacy recorder stamped each snapshot with request-start/rotation time before its blocking REST fetch completed. Frozen V2 replay applied the returned book at that early tag. The current recorder persists request and response times separately. Under `post_response_proxy_depth_boundary_v1`, replay emits a fail-closed barrier at recorded local request time, then establishes a valid bridge and requires a sequence-valid retained depth receipt after response completion. Legacy files use the first strictly later receipt as a post-response upper-bound proxy. Reconstruction is released at the selected record's `E`, and only its final state reaches strategy logic. Local request time, depth `E`, trade `T`, and receipt times retain different semantics and uncalibrated latency. This is not a calibrated observation clock.

Perpetual-futures depth and raw trade capture also exists, isolated in separate directories so its schema cannot collide with spot. That dataset remains recording-only: this report makes no spot-perpetual comparison.

3.2 Integrity inventory and panel selection

The frozen manifest inventories 1193 hourly slots before 1 June 2026. Of these, 775 passed the required file, gzip, JSON, snapshot-bridge, and sequence checks, 418 did not. Most failures reflect missing recorder files, with some decode or sequence issues. Any five-hour window containing an invalid hour was ineligible.

Development selection retained the six exploratory anchors and filled the panel with non-overlapping valid windows while balancing UTC buckets and one-second realized-volatility terciles. The result is 24 development windows and 12 holdout windows. Development is concentrated in 12–20 April and 6–9 May. The holdout is in late May. Clean recording is therefore not assumed to be missing completely at random. If feed or recorder failure correlates with stressed markets, eligibility can itself select the sample.

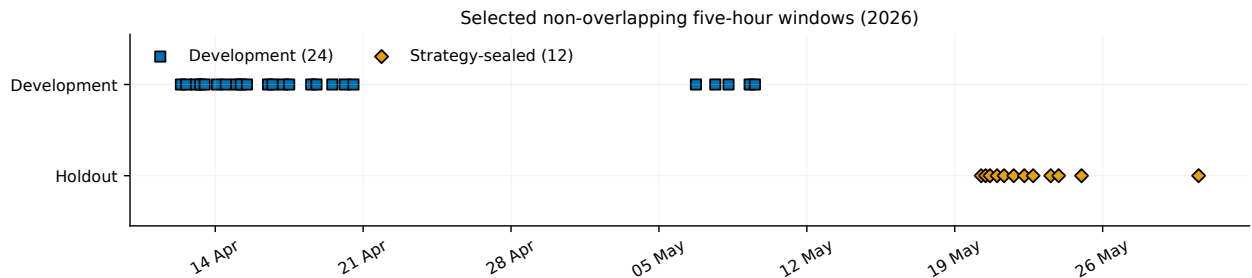


Figure 1: Frozen selected panel. The holdout is strategy-sealed, not unseen: window identities and pre-strategy descriptors were examined, but no candidate strategy was evaluated on them. The regime comparison labels the holdout *regime-shifted*, principally because realized volatility and jump counts were lower.

The full integrity identity begins `a3a99b0a616a`. The selected panel identity begins `760c55b7c092`. Exact SHA-256 values and every selected start are committed in the canonical artifacts.

A publication-time audit hash-checked all 120 selected development depth files. Every snapshot bridged. The first later local depth receipt followed the legacy tag by a median 250.5 ms, 90th percentile 672.9 ms, and maximum 2,335 ms. This is an upper-bound response-completion proxy, not a measured round trip. Frozen fills associated with orders placed before the selected policy boundary numbered 2 at $c = 0$ and 4 at $c = 1$. This is a proxy diagnostic, not an upper bound or a robustness result, and it cannot exclude changed queue age. In all 120 files the selected policy boundary was the valid bridge, but its replay time remains exchange E. The full V3 panel must be rerun under the new policy.

3.3 Data availability

Raw recordings are not stored in Git. At the report date the complete local multi-dataset capture is approximately 37 GB, including unused perpetual-futures data. The two spot directories used here occupy approximately 20.5 GB. The 240 selected development files occupy approximately 0.82 GB. The tracked repository contains only selected derived artifacts. This gives three distinct reproducibility levels:

1. a fresh clone can run deterministic unit tests and a synthetic event-driven lifecycle demo,
2. a fresh clone can validate the committed artifact structure, expected identities and verdict fields, regenerate report figures, and confirm that no candidate holdout artifact exists, and
3. reproducing replay-derived numbers requires the locally held raw files whose hashes are recorded in the integrity manifest.

The repository therefore uses the precise term *artifact-verifiable*. It does not claim that the full empirical panel can be rerun from a clone alone.

3.4 End-to-end architecture

Figure 2 shows the main information flow. Strategies express intent through order or cancellation requests. The execution simulator, not the strategy, owns arrival time, order state, queue position, fills, and fees. This boundary keeps trading logic independent of execution mechanics and makes it possible to test the latter with synthetic event paths.

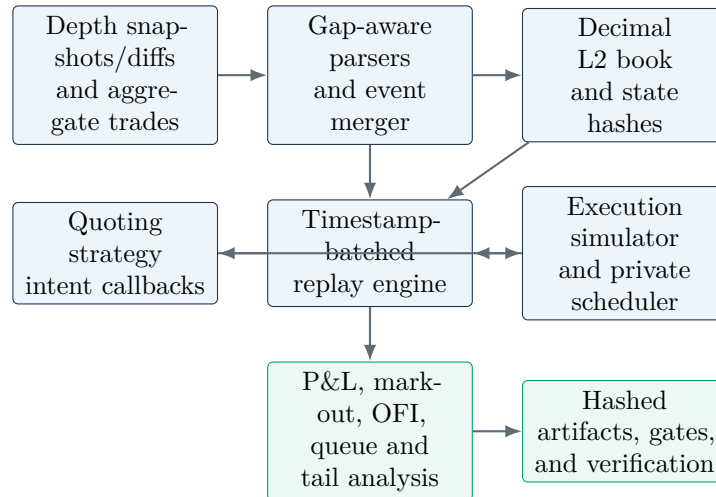


Figure 2: System architecture. Recorded market data and simulated private actions meet on the deterministic modeled clock defined in Section 4.

3.5 Component responsibilities

Table 3: Primary code boundaries.

Component	Responsibility
<code>src/replay/orderbook.py</code>	Sorted aggregate depth, best prices and sizes, mid, spread, microprice, snapshots/diffs, deterministic state hash.
<code>src/replay/depth_parser.py</code>	Hourly depth decoding, snapshot bridging, stale-diff rejection, and sequence-gap propagation across files.
<code>src/replay/trade_parser.py</code>	Aggregate-trade decoding, modeled ordering field, aggressor side, and ID-gap detection.
<code>src/replay/event_merger.py</code>	Stable two-way market-data merge and the recorded depth-before-trade tie rule.
<code>src/replay/engine.py</code>	Timestamp grouping, book mutation, strategy callbacks, private-action scheduling, gap policy, and replay termination.
<code>src/execution/simulator.py</code>	Entry/cancel latency, order lifecycle, post-only checks, FIFO queue state, partial fills, volume conservation, and fees.
<code>src/strategies/base_mm.py</code>	Inventory and live-order tracking, tick rounding, requote behavior, and the hard working-exposure envelope.
<code>src/analysis</code>	Markouts, P&L decomposition, matched lots, bootstrap intervals, OFI/microprice tests, fee and queue sensitivity, and tail diagnostics.

3.6 Determinism

Prices, quantities, fees, queue ranks, and accounting use `decimal.Decimal`. Latency jitter uses seeded independent random streams for entries and cancels. Replay uses recorded timestamps without runtime clock reads. Every checkpoint hashes a canonical serialization of full book state. Determinism is necessary for debugging, regression testing, artifact identity, and eventual cross-language parity. It does not by itself validate the economic assumptions of the simulator.

4 Execution-Model Contract

4.1 Why execution is part of the estimand

For a passive maker, the fill set is generated jointly by market events and the simulator’s rules. Changing arrival timing, cancellation timing, queue advancement, or post-only handling can change which trades fill an order. It can therefore change P&L, markouts, conditional signal tests, and apparent latency sensitivity. Execution assumptions are not an implementation footnote. They define the sample on which the strategy is evaluated.

L2 data reports aggregate quantity at a price, not individual order identities. No replay using this dataset observes true queue rank. The implementation is a documented execution approximation, not an exchange twin.

4.2 Historical and current semantics

Table 4: Execution-model boundary.

Feature	legacy_book_update_v1	event_driven_v2
New-order arrival	Eligible on the next depth update after modeled arrival	Scheduled at the exact arrival millisecond on the model clock
Cancellation	Effective immediately when requested	Scheduled separately, order stays fillable while cancel is in flight
Cancel/fill race	Absent	Explicit, a fill can beat a late cancel
Equal-time private action	Implicit through depth activation	All recorded market data at the millisecond precedes private arrivals
Own orders at one price	Incomplete separation of public and own queue	Non-overlapping own FIFO ranks with trade-volume conservation
Gap state	Strict replay policy added, but legacy artifacts retain old private semantics	Whole timestamp group invalidated fail-closed until snapshot resync
Snapshot timing	Returned state applied at a pre-fetch local tag	Local request-time resync barrier, recovery gated by post-response receipt order and released on depth E, new captures store request and response times
Artifact root	results/panels/btcusdt_12_panel_v2	results/panels/btcusdt_12_panel_v3_event_driven
Research status	Frozen historical evidence	Mechanics accepted, economic rerun pending

4.3 Clock and equal-timestamp policy

The replay clock has two classes of event. Its recorded-data axis combines local request time for fail-closed resync barriers, depth E for ordinary diffs and recovered-state release, and trade T for trades:

1. recorded market data, ordered depth before trade for unresolved millisecond ties and preserving source order within each stream, and
2. simulated private arrivals for new orders and cancellations, preserving stable insertion order.

Private actions strictly before a market-data timestamp are processed first. At an identical millisecond, the complete recorded market-data group is processed before private arrivals. A new order arriving at the same millisecond as a trade is therefore not eligible for that trade. A cancel arriving at the same millisecond loses the race. This is a conservative policy for unresolved timestamps, not a statement about the venue’s internal matching sequence.

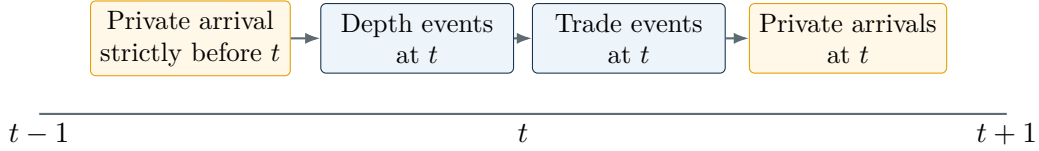


Figure 3: `event_driven_v2` timestamp policy. The horizontal placement inside the t group shows processing precedence, not sub-millisecond observation.

Depth-decrease attribution is deferred until every trade at the same millisecond has been observed. Otherwise the depth-before-trade rule could mistake traded quantity for a cancellation and grant queue improvement twice.

4.4 Order and cancellation lifecycle

Submitting an `OrderRequest` creates a pending order and schedules its arrival at

$$t_{\text{arrive}} = t_{\text{submit}} + L_e + J_e, \quad (1)$$

where L_e is base entry latency and J_e is a seeded uniform integer jitter bounded so that delay cannot be negative. Cancellation uses a separate latency and independent random stream:

$$t_{\text{cancel}} = \max(t_{\text{request}} + L_c + J_c, t_{\text{arrive}}). \quad (2)$$

While entry is pending, its quantity is counted as potential exposure but it is not eligible for market fills. Once active, the order remains fillable until the cancellation reaches the simulated exchange. Lifecycle events distinguish request from effect: `cancel_requested`, `cancelled`, and `cancel_too_late`. A post-only limit that would cross on arrival is cancelled with an explicit reason rather than silently reclassified as a passive fill.

4.5 Queue model and own-order FIFO

When a resting order arrives, its modeled quantity ahead is decomposed as

$$Q_i^{\text{ahead}} = Q_i^{\text{external}} + Q_i^{\text{own}}. \quad (3)$$

The external component starts from displayed L2 quantity, subject to a FIFO floor imposed by earlier live own orders. The own component is the remaining quantity of earlier simulated orders at the same side and price. A public depth decrease not explained by recorded trades improves only the external component:

$$Q_{i,t+1}^{\text{external}} = Q_{i,t}^{\text{external}} (1 - c\rho_t), \quad c \in [0, 1], \quad (4)$$

where c is queue-cancellation credit and ρ_t is the inferred fraction of the displayed queue removed without matching trade volume. The two historical endpoints are $c = 0$ and $c = 1$. Neither is claimed to be calibrated ground truth.

Recorded trade quantity advances non-overlapping FIFO ranks. The simulator plans and validates the complete transition before mutating any order and asserts

$$\sum_i q_{i,t}^{\text{maker fill}} \leq q_t^{\text{recorded trade}}. \quad (5)$$

Cancelling an earlier own order releases its unfilled quantity from later own queues. Decimal comparisons use only a scale-aware 10^{-24} relative tolerance for division residue. The tolerance cannot create fill budget.

4.6 Risk envelope

Position is not the only relevant exposure during cancel/replace. If multiple same-side entries are pending or cancels are in flight, all may still fill. The strategy therefore checks worst-case working exposure:

$$\left| I_t + \sum_{i \in \mathcal{F}_t^{\text{same}}} s_i q_i^{\text{remaining}} + s_{\text{new}} q_{\text{new}} \right| \leq I_{\max}, \quad (6)$$

where $\mathcal{F}_t^{\text{same}}$ includes active orders, pending entries that may become fillable, and active orders with cancels in flight. An invariant breach aborts replay. It is not recorded as a harmless temporary overshoot.

4.7 Gaps and replay termination

A known sequence gap invalidates the entire timestamp group before any event in that group can fill an order or invoke strategy logic. Local orders and pending actions become invalid because their queue state is no longer trustworthy. Replay resumes only at a valid snapshot. This is fail-closed research bookkeeping, not a claim that the exchange cancelled those orders instantly.

Private events scheduled after the last recorded market event are not executed against an assumed future book. Remaining live orders are terminalized as expired, and pending action counts are reported separately.

4.8 Known execution limits

- Aggregate L2 cannot reveal true market-by-order rank, hidden liquidity, or where cancellations occurred relative to the simulated order.
- The modeled clock mixes recorded local request time for resync barriers, depth event time **E** for release and ordinary diffs, trade matching time **T**, and local receipt order used to gate snapshots. Those fields are not guaranteed to be perfectly comparable, and millisecond resolution cannot identify true within-millisecond matching order. The post-response receipt gate removes request-start placement, but its selected boundary remains on exchange **E**. It does not calibrate client observation or feed latency.
- Market-data latency is not modeled separately from private-message latency.
- Counterfactual taker execution shares a shadow liquidity ledger only until the next observed depth update, taker-heavy research needs market impact and stronger book reconciliation.
- No execution model here has been calibrated against live order acknowledgements and fills from an exchange account.

5 Research Design

5.1 Primary quantities

Let best bid and ask be b_t and a_t , with displayed top-level quantities q_t^b and q_t^a . The arithmetic mid and microprice are

$$m_t = \frac{b_t + a_t}{2}, \quad \mu_t = \frac{q_t^b a_t + q_t^a b_t}{q_t^b + q_t^a}. \quad (7)$$

Microprice deviation is $(\mu_t - m_t)/m_t \times 10^4$ bps. For horizon h , the population response is forward mid drift,

$$r_{t,h} = 10^4 \frac{m_{t+h} - m_t}{m_t}. \quad (8)$$

The top-of-book OFI increment follows the event construction in Cont et al. [3]:

$$e_n = \mathbf{1}_{b_n \geq b_{n-1}} q_n^b - \mathbf{1}_{b_n \leq b_{n-1}} q_{n-1}^b \quad (9)$$

$$- \mathbf{1}_{a_n \leq a_{n-1}} q_n^a + \mathbf{1}_{a_n \geq a_{n-1}} q_{n-1}^a. \quad (10)$$

Raw one-second OFI sums e_n over the preceding second. Normalized OFI divides that sum by mean displayed best-bid-plus-best-ask depth over the contributing updates. The code definition is in `src/analysis/of_i_signal.py`.

For a maker fill i , let $s_i = +1$ for a buy and -1 for a sell. The fill-conditioned response is

$$z_{i,h} = 10^4 s_i \frac{m_{i+h} - m_{i-}}{m_{i-}}, \quad (11)$$

where m_{i-} is the latest book sample strictly before the fill. This quantity isolates subsequent mid movement from transaction-price spread capture. The separate standalone markout diagnostic uses fill price as its reference and is not substituted into this premise test.

Strictly-pre-fill anchoring is enforced in both the OFI window and reference mid. A book sample stamped at the fill millisecond is excluded, because it can already contain the same-millisecond market move associated with the simulated fill. Tests contain a leakage tripwire for this boundary.

5.2 Baseline and development panel

The frozen baseline is a post-only `MicropriceMM` on Binance spot BTCUSDT. It quotes 0.001 BTC on each eligible side, 2 USDT from `microprice`, requotes at most every five seconds, and caps position at 0.01 BTC. Historical execution assumptions are 10 ms entry latency without jitter, a 2 bps maker fee, a 5 bps taker fee, and queue-cancellation credit evaluated at $c = 0$ and $c = 1$. Each nominal five-hour window is the sum of five independently initialized one-hour replay episodes. Strategy, execution, and inventory state restart each hour, residual inventory is marked to the session-end mid and then discarded without a modeled liquidation. The five-hour value is therefore an inference cluster, not continuous five-hour portfolio P&L. Latency and fees are modeled inputs, not measurements from the author's exchange account.

The 2 USDT / five-second baseline was selected during an exploratory April anchor analysis. Six such anchors were retained among the 24 development windows. The expanded panel therefore reduces dependence on those examples but is not an untouched confirmation sample. That qualification is important when interpreting the later confidence intervals.

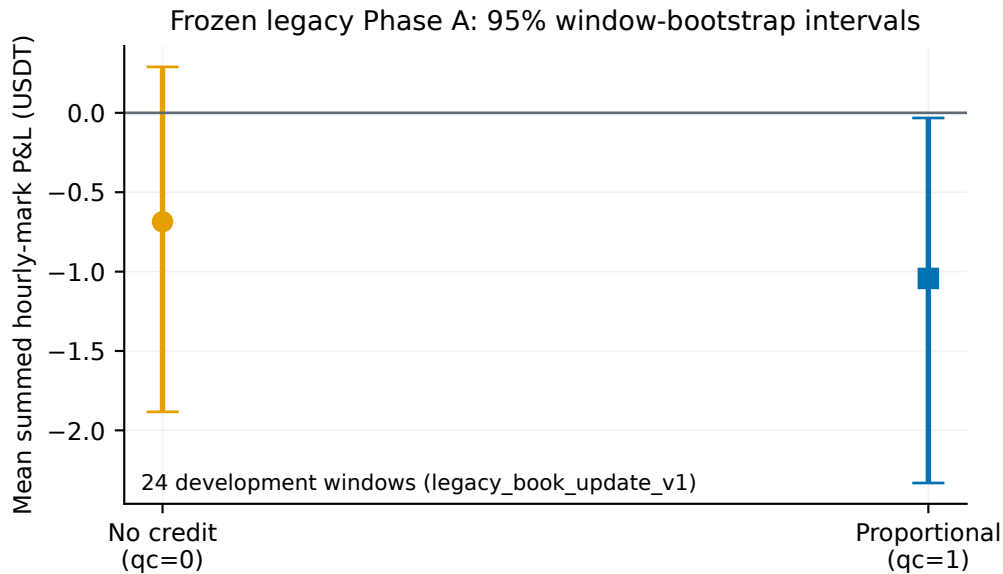


Figure 4: Phase A estimand: the mean full-strategy net P&L across 24 five-hour development windows. Intervals resample windows, not individual events.

5.3 Sequential decision protocol

The Phase A/B/C sequence and thresholds were written into the repository before their associated evaluation. This is a pre-specified protocol in the repository history. It is not an externally registered study.

1. **Phase A: baseline remeasurement.** Estimate mean net P&L and a 95% nonparametric interval by resampling the 24 five-hour windows 10,000 times with seed 7, separately at $c = 0$ and $c = 1$.
2. **Phase B: premise first.** Before running a strategy that acts on OFI, require a stable population association and a material fill-conditioned separation. If the premise fails, do not search the candidate's thresholds.
3. **Phase C: model stress.** Sweep queue-cancellation credit at $\{0, .25, .5, .75, 1\}$ under 10 ms latency, plus $\{0, 50\}$ ms at the two credit endpoints. Compare throughput, matched-lot economics, and full-strategy P&L.

The primary one-second normalized-OFI population gate required all of:

- one coefficient sign in at least 75% of development windows,
- pooled HAC $|t| \geq 2$, and
- at least 0.05 bps predicted drift for one signal standard deviation.

The conditional gate compared, at 30 seconds, the most-populated negative side-aligned bucket with the most-populated nonnegative bucket. It required the latter's mean $z_{i,30s}$ to exceed the former by at least 1.0 bps, with at least 30 observations in each selected bucket. The 1.0 bps threshold is a research materiality heuristic and 30 is a minimum-count rule. Neither is a formal power calculation. The chosen comparison also does not test the two extreme buckets. For those reasons the complete bucket response, its monotonicity, and its counts must accompany the mechanical gate result.

5.4 Dependence and uncertainty

Overlapping forward returns are serially dependent. Pooled regressions therefore report Newey–West heteroskedasticity-and-autocorrelation-consistent standard errors [6]. The implementation takes the larger of the overlap-horizon lag count and the Newey–West sample-size rule, capped only at $n - 1$. This corrects standard errors for the specified local dependence structure, but it does not turn millions of one-second observations into millions of independent market regimes. The pooled population regression does not yet carry window-clustered uncertainty across market regimes.

Phase A uncertainty instead resamples whole five-hour windows, preserving within-window dependence [4]. With only 24 clustered windows, the interval describes variation within this mechanically selected panel. It does not estimate sampling variation from a random population of market regimes. A future inference pass should add window-clustered uncertainty to both the pooled population effect and the fill-conditioned bucket contrast rather than interpreting their raw event counts as independent precision.

5.5 Holdout discipline

Twelve late-May windows were selected and their identities hashed alongside the development panel. This holdout is strategy-sealed rather than unseen. A pre-strategy regime screen labeled it **regime-shifted**. It had lower realized volatility and fewer jumps than development. The holdout protocol requires a candidate, parameters, gate result, and commit to be locked before one evaluation. Because the OFI premise gate blocked the only candidate, the template remains unlocked and no holdout strategy output exists. No candidate evaluation was run, so there is no favorable or unfavorable strategy result kept off the page.

6 Frozen Development Results

Provenance. Every number in this section belongs to the frozen V2 artifact root and historical execution model. These are results of that experiment, not estimates from the current simulator. V2 also applied REST snapshot state at a pre-fetch local tag. The timing proxy audit does not prove that its execution-derived values are invariant to the post-response-gated exchange-clock policy.

6.1 Phase A: passive baseline

The fixed microprice maker was unprofitable in 18 of 24 windows at both queue endpoints. Table 5 reports window means and 95% bootstrap intervals. Proportional cancellation credit produced a wholly negative interval. The no-credit interval crossed zero, but its point estimate was also negative. No stable passive edge was demonstrated. This conclusion retains the queue-model dependence of the uncertainty.

The fee hurdle is economically central. At a representative 71,500 USDT price, the 2 USDT half-spread is about 0.28 bps per side, while the simulation charges 2 bps per maker fill. In the pooled fee diagnostic, completed matched lots would break even at maker fees of 1.351 bps under no cancellation credit and 0.426 bps under proportional credit. Including the hourly residual-inventory marks, the full strategy would instead require maker rebates of 0.769 and 1.351 bps, respectively. The baseline is therefore a demanding execution diagnostic, not a configuration with an assumed spread advantage over fees.

The frozen fee summary metadata calls this a “window-close” mark. That phrase is a preserved wording error: the computation marks each one-hour session end and resets state before the next session. The numeric rows are unchanged.

Table 5: Frozen legacy Phase A full-strategy net P&L per nominal five-hour development window. Each row sums five independently initialized, hourly marked episodes (0.001 BTC quotes, 0.01 BTC maximum position, 2 bps maker fee).

Queue credit	Mean (USDT)	95% CI low	95% CI high
0 (no credit)	-0.686	-1.883	+0.290
1 (proportional)	-1.043	-2.332	-0.032

The original reference was microprice because it can encode top-of-book imbalance. Its own one-second signal was nevertheless weak: +0.0190 bps per signal standard deviation, HAC $t = 1.74$, with a positive coefficient in 19 of 24 windows. It missed both the $|t| \geq 2$ and 0.05 bps materiality gates. This motivated OFI as a richer event-flow signal without claiming that the baseline was optimized out of trouble.

6.2 Phase B: OFI population evidence

Normalized OFI passed all three population gates. The one-second coefficient was positive in 24 of 24 windows. Its unconditional bucket response rose monotonically from -0.276 bps in the lowest bucket to $+0.284$ bps in the highest. Table 6 shows that statistical significance persists at longer horizons while explanatory power falls sharply.

Table 6: Frozen legacy pooled normalized-OFI regressions on forward mid drift.

Horizon	Observations	Slope	HAC t	Effect / 1 s.d. (bps)
1 s	430,278	0.1192	64.63	0.1233
10 s	430,062	0.2424	35.83	0.2507
1 m	428,862	0.3157	18.06	0.3268
5 m	423,102	0.2958	7.48	0.3069

The one-second regression R^2 is only 0.0366. The large t -statistic reflects a repeatable short-horizon association in a densely sampled series. It does not imply that the signal explains most price variation or can be captured after execution and costs.

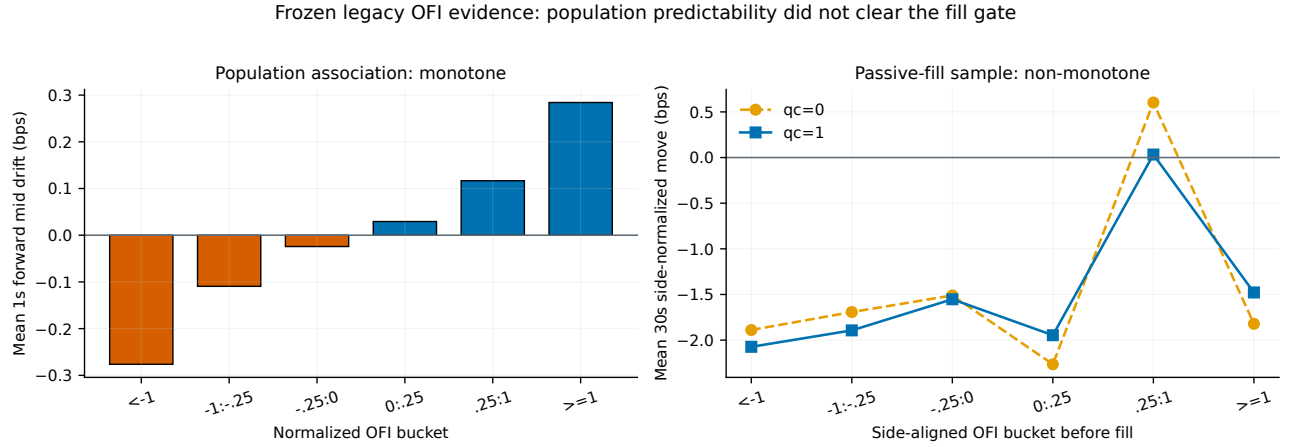


Figure 5: Unconditional and fill-conditioned OFI evidence. The left panel is queue-independent. The right panel uses historical baseline fills and must be interpreted under the legacy execution model.

6.3 Phase B: the conditional gate

The pre-specified 30-second conditional comparison was only +0.127 bps at $c = 1$ and -0.376 bps at $c = 0$, far below 1.0 bps. Its selected bucket counts were respectively 201 and 154, exceeding the minimum-count rule. The full response was not monotone. In particular, the smaller $[0.25, 1)$ bucket appeared more favorable than the selected adverse bucket at both endpoints, but had only 71 and 54 observations. That is an exploratory pattern, not a post-hoc replacement for the defined test and not a powered estimate of a tradeable threshold.

The defensible result is consequently precise: the defined fill-conditioned gate failed at both queue endpoints, and the broader bucket pattern was not stable or monotone enough to advance the candidate. It would be too strong to say that OFI’s predictive content universally “vanished” on fills, or that an unrun OFIGatedMM necessarily loses.

6.4 Phase C: queue and latency stress

At 10 ms entry latency, granting more cancellation credit raised modeled fill throughput but worsened both quantity-weighted matched-lot economics and total strategy P&L. This monotone economic deterioration is stronger evidence than the sign flip observed in the earlier six-anchor analysis.

Table 7: Frozen legacy pooled Phase C results at 10 ms. “Orders/fill” is a throughput diagnostic, matched net/BTC weights pooled matched quantity.

Credit	Fills	Orders/fill	Matched net/BTC	Full net P&L
0.00	1,595	70.8	−9.75	−16.46
0.25	1,796	62.9	−15.08	−17.64
0.50	1,802	62.6	−16.08	−20.88
0.75	1,831	61.6	−17.51	−21.33
1.00	2,041	55.4	−23.65	−25.03

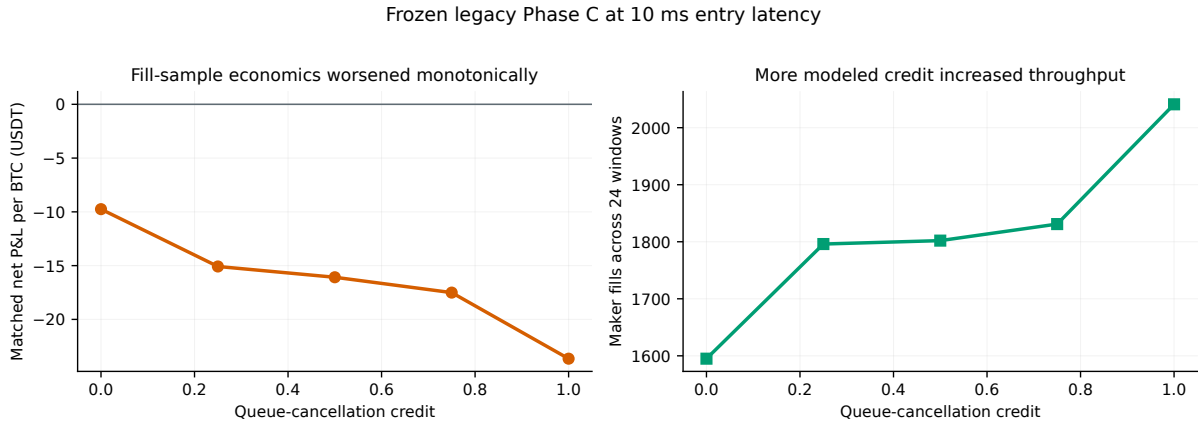


Figure 6: Queue-credit sensitivity. More favorable modeled queue advancement selects a larger pooled fill sample with worse mean matched economics. The sweep does not establish that fill sets are nested.

The historical 0/10/50 ms entry-latency sweep was numerically unchanged at the two credit endpoints. Under `legacy_book_update_v1`, order eligibility was quantized to the next depth update, so this is evidence about a model limitation rather than evidence that real latency is irrelevant. It was one reason to build the event-driven clock.

6.5 Tails and bounded replay audits

Under proportional credit, the worst 5% of 30-second fill observations accounted for 40.64% of the summed adverse movement. The worst 5% of matched lots accounted for 86.62% of total matched-lot loss. Fill toxicity was thus broad with meaningful tails, whereas realized matched-lot loss was far more tail-concentrated. These ratios depend on the signed-total denominator and are reported with negative-only denominators in the artifacts as well.

The panel-wide same-millisecond audit found 10 overlap fills among 2,041 fills at $c = 1$ (0.49%) and 7 among 1,595 at $c = 0$ (0.44%). It found zero *artifact-evidenced* contradictions of the depth-before-trade assumption. That bounded label matters: reconciliation artifacts do not persist every queue-drain timestamp, so candidate rows cannot all be adjudicated after the fact. Separately, all six original anchor windows were aggregate-trade-gap clean, and replaying them under the later fail-closed trade-gap policy changed no reported value.

6.6 Decision

No strategy advanced. The baseline did not demonstrate stable positive economics, microprice was weak, and OFI cleared the population screen but not the specified execution-conditioned premise. Phase C showed that reasonable variation in unobserved queue advancement changed throughput and loss

magnitude without producing positive pooled matched economics. No candidate strategy was evaluated on the holdout. This is a negative/conditional development result, not a live-performance claim.

7 Failure Analysis and Lessons

7.1 What changed after apparent success

The project is most informative when read as a sequence of claims that became more constrained as the measurement system improved.

Table 8: Material failure modes and their disposition.

Issue	Why it mattered	Resolution
Accidental taker fills	Entry latency could make intended passive limits cross, creating attractive but irrelevant execution.	Post-only is the default, crossing arrivals terminate with an explicit reason.
Naive snapshot synchronization	Stale post-snapshot diffs could roll the reconstructed book backward.	Stale diffs are dropped, the first retained diff must bridge, and parser tests cover both rules.
Pre-fetch snapshot tags	The recorder stamped request start, then blocked on REST, legacy replay could apply returned state before response completion.	Frozen V2 is labeled, current recovery is gated by post-response receipt order, future captures store both timestamps, and a 120-hour proxy audit is committed.
Trade-ID gaps	Missing aggressor trades can overstate queue position and suppress fills.	New research pauses fail-closed, invalidates local state, and resumes only at a valid snapshot, six anchors were audited before/after.
Same-ms fill leakage	A fill-time book sample could include the causal trade and leak its move into prior OFI.	Conditional OFI and its reference mid are strictly before the fill, a test guards the boundary.
Private-event quantization	Legacy entry arrival waited for a depth event and cancels had no delay, masking races and latency sensitivity.	Historical results were frozen and labeled, <code>event_driven_v2</code> added explicit millisecond scheduling and a new artifact namespace.
Own-order overlap	Multiple simulated orders could compete for the same recorded trade or obscure working exposure.	Current replay separates public and own queue, conserves trade volume, models own FIFO, and checks all fillable same-side quantity.
Publication overstatement	Phase 1 ratios and the OFI headline could imply stronger evidence than the implementation supports.	Phase 1 is now an educational precursor, the OFI conclusion is stated as a failed defined gate with non-monotone conditional evidence.

When a correction could change the estimand, the project did not silently overwrite the interpretation. Contaminated outputs were superseded, corrected studies were regenerated, and the private-timing change created an explicit model version. That separation is essential for an auditable research record.

7.2 Empirical limitations

- **One venue and instrument.** Evidence comes from Binance spot BTCUSDT and cannot establish cross-venue or cross-asset robustness.
- **Short, clustered period.** Development windows lie mainly in two April/May clusters, calendar, volatility, and market-structure coverage is limited.
- **Selective availability.** Only 775 of 1,193 inventoried hours passed the data requirements. Missingness may correlate with volatile or operationally difficult periods.
- **Exploration inside development.** Six baseline-selection anchors remain in the 24-window panel, results are not a pristine confirmatory replication.
- **Hourly state resets.** Each five-hour value sums five independent one-hour episodes. Residual inventory is marked at each session end and discarded without liquidation cost, so the result is not continuous five-hour portfolio P&L.
- **Strategy-sealed but shifted holdout.** The 12 late-May windows are lower volatility and have not been used for a candidate.
- **Limited conditional inference.** The OFI bucket minimum is not a power analysis, the defined contrast is not an extreme-to-extreme comparison, and a window-clustered interval for that contrast remains future work.
- **Pooled-signal inference.** Newey–West errors address specified local serial dependence, not uncertainty across independently sampled market regimes. A window-clustered population-effect interval remains future work.

7.3 Model limitations

Aggregate L2 supplies displayed quantity, not order identities. The true queue rank of a hypothetical order is unobservable, and cancellation credit brackets one uncertainty without calibrating it. Hidden liquidity, iceberg refresh, self-trade prevention, exchange throttles, acknowledgement latency, network jitter, message loss, and venue-specific matching details are absent. The millisecond tie rule is deterministic but cannot recover sub-millisecond truth.

The simulator assumes the recorded public market remains unchanged by the strategy. That is reasonable only for very small passive orders. It is not a market-impact model, a capacity study, or a production matching-engine replica. The 2 bps maker fee is a fixed assumption rather than an observed account-tier fee. It is also large relative to the 2 USDT half-spread at the observed BTC price level, making fee calibration load-bearing.

7.4 Software and operational limits

The current Python implementation is a tested behavioral reference. It favors explicit state and `Decimal` arithmetic over throughput, has not been profiled as a low-latency system, and is not connected to an order gateway. Tests cover deterministic synthetic paths and artifact contracts. They do not replace exchange conformance tests, live shadowing, chaos testing, a production gateway, kill switch, monitoring, or other operational risk controls.

Raw data is not distributable from this repository, so a fresh clone can verify committed evidence but cannot independently reconstruct every economic number. The data statement records this boundary and the expected local layout.

7.5 Technical lessons

Five lessons carry forward:

1. execution determines the sample, so a predictor must be tested on the fills it can actually influence,

2. a simulator assumption should be versioned whenever it can change order eligibility or queue state,
3. negative results become useful when their stop rule and provenance are inspectable,
4. sensitivity analysis should report selection effects, including throughput and fill composition, alongside the final P&L number, and
5. performance work should begin only after a deterministic behavioral contract exists to serve as an oracle.

These are the project’s principal learnings. They are more durable than the economic estimates from one narrow historical panel.

8 Current Status and Reproduction

8.1 Where the project is now

Table 9: Publication-time status.

Workstream	Status	Meaning
Bar backtester	Educational precursor	Reusable framework and tests remain, historical performance claims are not treated as rigorous evidence.
Frozen L2 study	Complete under legacy model	Phase A/B/C artifacts, gates, panel identity, and audits are committed and verifiable.
Event-driven Python model	Mechanics accepted	Private scheduling, post-response-gated snapshot recovery, delayed cancels, own FIFO, conservation, gaps, and risk invariants pass deterministic tests.
V3 development economics	Pending	No execution-derived result from <code>event_driven_v2</code> is published yet.
Holdout	Strategy-sealed	Identities and regime descriptors were examined, but no candidate strategy run exists.
C++ performance port	Intentionally paused	It begins only after the Python V3 result is frozen and a separate modern C++ foundations phase is complete.

The immediate research milestone is the clean V3 development rerun under `event_driven_v2` and `post_response_proxy_depth_boundary_v1`, including an explicit before/after snapshot-timing comparison. It is not the C++ port and it is not a holdout run.

8.2 Fresh-clone verification

The repository targets Python 3.11. At the 4 August 2026 checkpoint, a fresh clone reports 376 deterministic tests passed and three optional raw-data smoke checks skipped. All 379 pass when the selected captures are restored. From the repository root:

```
python3.11 -m venv .venv
source .venv/bin/activate
python -m pip install -r requirements.txt

env PYTHONPATH=. pytest -q
env PYTHONPATH=. python scripts/demo_event_driven_execution.py
```

```
env PYTHONPATH=. python scripts/verify_v2_artifacts.py
```

The synthetic simulator demo is deliberately tiny: a resting buy arrives, a cancellation is requested, a recorded sell reaches the queue first, and the delayed cancel is reported too late. It exercises execution lifecycle mechanics without raw data. It does not exercise full replay timestamp batching, gap recovery, empirical economics, or throughput.

The V2 verifier checks the manifest and panel identities, execution-model marker, Phase A verdict, Phase B gate status, Phase C grid, same-millisecond audit, panel-file hashes, and selected holdout identifiers across tracked result paths and contents. It validates committed evidence. It does not recompute replay results from raw messages.

8.3 Building this report

Report scalars and vector figures are regenerated only from an allowlist of canonical frozen artifacts. The builder checks the execution-model marker and input shape, including the distinction between 24 window rows and one pooled row per queue/latency configuration.

```
make report
make report-check
```

The first command rebuilds report inputs and the PDF. The second rebuilds and then verifies that the generated asset manifest matches the committed copy. Input/output SHA-256 values are in `report/generated/asset_manifest.json`. The compiled document is `report/l2_mm_research_report.pdf`.

8.4 Full development rerun

With the raw depth/trade layout described in `DATA.md`, the current development runner defaults to the frozen development-window file and writes only to the V3 namespace:

```
env PYTHONPATH=. python scripts/run_l2_panel.py --phase all
```

Before relying on this command, use `-help` and inspect the committed defaults. The runner verifies the development-panel file hash, refuses the strategy-sealed holdout, and guards the frozen V2 directory. V3 artifact verification remains impossible until this run produces its manifest.

8.5 Suggested reviewer routes

A reviewer can understand the project at three depths:

Five minutes

Read the repository README, the executive summary, and the status table above.

Thirty minutes

Read Sections 3–7 and run the synthetic demo plus V2 artifact verifier.

Code review

Start at `src/replay/engine.py` and `src/execution/simulator.py`. Use `tests/test_execution_simulator.py` and `tests/test_engine.py` as executable specifications, then inspect `src/analysis/of_i_signal.py` for the research estimand.

8.6 Roadmap to the performance boundary

1. Run the fixed baseline over all 24 development windows under `event_driven_v2` and the post-response-gated snapshot policy at both queue endpoints.
2. Regenerate OFI conditional-on-fill evidence and queue/latency stress, then compare fill identity and economics with the legacy study.

3. If a candidate clears a newly committed development rule, lock it and perform the one-shot holdout strategy evaluation. Otherwise close the research arc without evaluating a candidate there.
4. Freeze a deterministic Python reference, golden event traces, artifact schemas, and parity tolerances.
5. Complete a separate modern C++ foundations phase before implementing any hot-path port. Later performance work must reproduce the Python contract before it is benchmarked.

8.7 Conclusion

The repository does not present a profitable market-making strategy. It presents something more credible at its current stage: a tested microstructure research system, a frozen negative experiment with auditable stopping logic, and a documented correction from coarse private timing to an event-driven execution reference. The remaining scientific question is whether the development conclusions survive the current model. The project stops at that boundary before any C++ performance implementation.

A Canonical Configuration

Table 10: Frozen V2 baseline configuration.

Field	Value
Venue / instrument	Binance spot BTCUSDT
Strategy / reference	MicropriceMM / top-of-book microprice
Windowing	24 development windows, each sums five independently initialized one-hour sessions
Half-spread / order quantity	2.00 USDT / 0.001 BTC
Maximum position	0.01 BTC
Requote interval	5,000 ms
Entry latency / jitter	10 ms / 0 ms
Fees	maker 2 bps, taker 5 bps
Queue cancellation credit	endpoints 0 and 1, Phase C grid 0, .25, .5, .75, 1
Market-data tie rule	depth before trade for identical millisecond timestamps
Legacy snapshot time	pre-fetch local tag, removed from strategy ordering only in the current model
Execution model	legacy_book_update_v1
Frozen research commit	1066950
Integrity manifest	a3a99b0a616a (full logical identity in artifact)
Selected panel	760c55b7c092 (full panel identity in artifact)

B Current Execution Acceptance Matrix

Table 11: Key `event_driven_v2` acceptance properties.

Property	Status	Executable evidence
Exact new-order scheduling	Accepted	Private arrivals occur at modeled milliseconds, including times with no depth update.
Delayed cancellation	Accepted	Request/effect are distinct and a fill can beat a cancel in flight.
Equal-time policy	Accepted	Full market-data group precedes private arrivals, insertion order is stable.
Snapshot recovery gate	Accepted	A local request-time barrier pauses replay, a sequence-valid post-response receipt gates release on selected depth E, strategy sees only the final recovered state.
Own-order FIFO	Accepted	Earlier own quantity forms non-overlapping rank ahead of later own orders.
Trade-volume conservation	Accepted	Aggregate maker fill cannot exceed recorded aggressor quantity.

Property	Status	Executable evidence
Queue-credit isolation	Accepted	Public inferred cancellation cannot erase own quantity ahead.
Hard exposure invariant	Accepted	Active, pending, and cancel-in-flight quantities are counted before quote placement.
Gap handling	Accepted	Gap timestamp group is invalidated and replay pauses until snapshot resynchronization.
Replay end	Accepted	Live orders expire and pending entries/cancels are reported.
Artifact separation	Accepted	Current model is guarded against writes beneath the frozen V2 root.
Development-panel economics	Pending	No V3 execution-derived results are published.

The detailed model contract is `notebooks/execution_model_v2.md`. Implementation tests are indexed under `tests` and grouped by behavior.

C Claim-to-Evidence Map

Current implementations have evolved since the frozen study. For historical claims, the implementation links below therefore point to the full frozen research commit. Current links are used only for later audits and verifiers.

Table 12: Fast audit map for principal claims.

Claim	Canonical artifact	Frozen implementation or verifier
Data eligibility and identity	<code>results/panels/btcusdt_l2_panel_v2/integrity_manifest.json</code>	<code>scripts/build_l2_integrity_manifest.py</code>
Development/holdout selection	<code>results/panels/btcusdt_l2_panel_v2/window_selection_summary.json</code>	<code>scripts/select_l2_windows.py</code>
Legacy execution boundary	<code>results/panels/btcusdt_l2_panel_v2/EXECUTION_MODEL.json</code>	<code>src/execution/provenance.py</code>
Phase A interval	<code>results/panels/btcusdt_l2_panel_v2/phase_a_verdict.json</code>	<code>scripts/bootstrap_baseline_ci.py</code>
Fee hurdle	<code>results/panels/btcusdt_l2_panel_v2/fee_break_even/btcusdt_microprice_hs2.00_rq5000_panel24/summary.json</code>	<code>scripts/analyze_fee_break_even.py</code>
Microprice premise	<code>results/panels/btcusdt_l2_panel_v2/microprice_signal/btcusdt_20260412_09_to_20260509_13_24windows_1000ms/summary.json</code>	<code>src/analysis/microprice_signal.py</code>

Claim	Canonical artifact	Frozen implementation or verifier
OFI population and fill gate	results/panels/btcusdt_l2_panel_v2/of_i_signal/btcusdt_microprice_ofi_20260412_09_to_20260509_13_24blocks_1000ms/summary.json	src/analysis/of_i_signal.py
Queue-credit stress	results/panels/btcusdt_l2_panel_v2/queue_credit_sweep/summary/summary.json	src/analysis/queue_credit_summary.py
Tail concentration	results/panels/btcusdt_l2_panel_v2/tail_diagnostics/btcusdt_microprice_hs2.00_rq5000_panel24/summary.json	src/analysis/tail_diagnostics.py
Same-ms audit	results/panels/btcusdt_l2_panel_v2/same_ms_audit/btcusdt_microprice_hs2.00_rq5000_panel24/summary.json	scripts/audit_same_ms_attribution.py
Snapshot-time exposure	results/replay_correctness/snapshot_timing_panel24.json	scripts/audit_snapshot_timing.py
No candidate holdout strategy result	notebooks/holdout_protocol.md	scripts/verify_v2_artifacts.py
Report presentation	report/generated/asset_manifest.json	scripts/build_report_assets.py

D Repository Map

src/replay	book reconstruction, parsers, event merger, replay clock,
src/execution	order lifecycle, queue model, fills, fees, provenance,
src/strategies	baseline and gated market-making strategies,
src/analysis	estimands, accounting, diagnostics, uncertainty, gates,
scripts	record, replay, run panels, analyze, verify, and build report assets,
tests	deterministic unit, invariant, integration, and artifact-contract tests,
results/panels	frozen selected derived artifacts, not raw market data,
notebooks	protocols, research log, historical writeups, and model notes,
report	this source, generated figures/macros, and compiled PDF.

E Terminology

Artifact-verifiable	A fresh clone can check a committed result’s schema, identity, and decision contract without recomputing raw replay.
Development	The 24-window panel on which baselines, premises, and model sensitivities were evaluated.

Holdout	Twelve strategy-sealed windows reserved for one locked candidate, identities and regime descriptors were inspected, but no candidate has been evaluated.
Matched net/BTC	Sum of matched-lot net P&L divided by sum of matched quantity, pooled rows are quantity-weighted, not averages of windows.
Population signal	Association measured over regularly sampled book states, whether or not the strategy fills.
Fill-conditioned signal	Association measured only for book states selected by historical passive fills.
Queue credit	Fraction of inferred public non-trade depth reduction credited as advancement ahead of the simulated order.

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